

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=yogurt_dessert

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	yogurt_dessert	silpo_novus	promo_controlled	nan	nan
prozorro	nan	nan	nan	nan	nan	nan	ambiguous	0.0	yogurt_dessert	silpo_novus	promo_controlled	nan	nan
retail	5.695496839419695e-10	0.1	1.468661222075579e-08	0.1	1.0051134444285687e-05	0.1	I(0)	0.0	yogurt_dessert	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	yogurt_dessert	silpo_novus	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	yogurt_dessert	silpo_novus	promo_controlled	nan	0.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	yogurt_dessert	silpo_novus	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
yogurt_dessert	silpo_novus	promo_controlled	daily	none	none	n/a	n/a	59	nan	nan	nan	nan	nan	nan	nan	unavailable	1	No admissible model estim...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
none	none	n/a	n/a	nan	nan	nan	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized	type	retailer	promo	variant	combined	rule
2025-10-21	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-22	nan	nan	51.39	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-23	nan	nan	54.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-24	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-25	nan	nan	52.9	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-26	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-27	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-28	nan	nan	52.9	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-29	nan	nan	52.695	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-30	nan	nan	52.9	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-10-31	nan	nan	52.49	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-01	nan	nan	52.695	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-02	nan	nan	52.49	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-03	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-04	nan	nan	50.74	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-05	nan	nan	51.44	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-06	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-07	nan	nan	46.24	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-08	nan	nan	50.74	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-09	nan	nan	52.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-10	nan	nan	46.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-11	nan	nan	44.89	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-12	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-13	nan	nan	52.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-14	nan	nan	46.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-15	nan	nan	55.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-16	nan	nan	49.4675	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-17	nan	nan	56.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-18	nan	nan	54.9	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-19	nan	nan	52.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-20	nan	nan	37.24	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-21	nan	nan	54.695	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-22	nan	nan	56.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-23	nan	nan	42.49	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-24	nan	nan	55.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-25	nan	nan	53.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-26	nan	nan	54.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-27	nan	nan	54.99	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-28	nan	nan	41.365	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			
2025-11-29	nan	nan	146.48250000000002	yogurt_dessert	silpo_novus	promo_controlled	daily_median_of_available...			

Table: LagProfile

No rows

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
yogurt_dessert	silpo_novus	promo_controlled	daily	ambiguous	ambiguous	I(0)	0	0	59	0	0	0	Low pair overlap in main ...

Table: NARDL_Multipliers

No rows

Table: VECM_IRF

No rows

time_series_promo_controlled.png

yogurt_dessert | silpo_novus | promo_controlled

